ODP6: SLF Crushers

Alice Park, Connor Hyde, Ei Thiri Kyaw, Nayana Zachariah, Serkan Yurday

1 Context and Problem Statement

Spotted lanternfly (SLF) infestations pose a major threat to vineyards, especially just before harvest when grapes are most vulnerable. SLF feeds on vine sap, causing yield losses of 30–50% in severe cases, along with increased labor and pesticide costs. We focus on removing adult SLF along the vines, as existing methods, such as shaking, suction, brushing, and high-force airflow, are unsuitable near ripe grapes due to the risk of fruit damage. To address this, we designed a non-destructive mechanical trap to attract, contain, and remove SLF without harming the crop. This prototype prioritizes low-cost, minimal-labor operations through a feasible trapping mechanism that uses low-power rotation to guide insects into a collection area. The use of captured live bodies naturally attracts more SLF into the trap, allowing fewer adults to target the grapes when traps are stationed along the rows. Furthermore, the design prioritizes drawing SLF away from grapevines to prevent reduced sugar content and long-term vine damage or death.

2 Final Prototype and Application

Our team developed a rotating trap designed to attract, guide, and capture spotted lanternflies (SLF) in a single integrated system. The device uses tree-of-heaven sap initially as an attractant to lure SLF onto the trap surface. Once on the device, slow rotational motion directs the insects toward a central opening, where they fall into a contained collection chamber below. As seen in Appendix A, the prototype is easy to assemble as you insert the D-shaft through the peg plate, leaving about 3 inches pushed in above. The motor-shaft connector goes above, connecting the shaft to the 3V motor. These joined pieces slide into the peg plate such that there is excess shaft poking below. The rotating floor is pushed in from under. Lastly, the collection chamber locks into the peg plate. The trap is designed for use in agricultural settings such as vineyards. It can be placed near affected plants and operate continuously with minimal human intervention. Since NYS farms prefer minimal human involvement in device operation, we designed our product to require human interaction only for emptying the container when it is full. The collected insects are stored in a removable chamber that can be emptied as needed, reducing the need for frequent pesticide applications and manual removal.

3 Results

3.1 Test Methodology and Success Criteria

Three subsystem tests evaluated the prototype's core functions, each targeting a distinct mechanical component. Performance is quantified using weighted indices that penalize failure modes by severity. (For more detailed information, refer to the appendix)

1. **Peg Guidance Test (GPI $\geq$ 80, high priority):** Assesses whether the reaper wall and peg plate guide a surrogate insect (marble) to the center opening over 20 trials. Penalties apply for bypass (B), jamming (J), and premature compression (C):

$$\mathrm{GPI} = 100\,(S_t - 0.4B_t - 0.7J_t - 1.0C_t)$$

   Score: 93.0 (19/20 successes, 1 bypass), exceeding the threshold.

2. **Chamber Drop Test (CRI $\geq$ 80, high priority — live demo):** Evaluates whether marbles/Orbeez enter the chamber cleanly over 20 trials. Penalties apply for partial entry (P), bounce-out (B), and missed opening (M):

$$\mathrm{CRI} = 100\,(S_c - 0.5P_c - 0.7B_c - 1.0M_c)$$

Score: 92.5 (19/20 successes, 1 partial entry), exceeding the threshold.

3. **Rotational Test (RSI $\geq$ 80, medium priority):** Evaluates whether the system completes 20 powered 360° rotations. Penalties apply for sticking (K), base displacement (D), and incomplete rotation (F):

$$\mathrm{RSI} = 100\,(S_r - 0.4K_r - 0.6D_r - 1.0F_r)$$

Score: 85.0 (18/20 successes, 1 sticking, 1 displacement), exceeding the threshold.

3.2 Results and Evaluation

Table 1 summarizes trial outcomes and index calculations (20 trials per test). Detailed criteria and methodology are provided in the Appendix.

Test	S	$B/P/K$	$J/B/D$	$C/M/F$	**Score**
Peg Guidance	$19/20$	$1/20$ (B)	$0/20$ (J)	$0/20$ (C)	$\mathrm{GPI} = 100(0.95 - 0.4(0.05)) = 93.0$
Chamber Drop	$19/20$	$1/20$ (P)	$0/20$ (B)	$0/20$ (M)	$\mathrm{CRI} = 100(0.95 - 0.5(0.05)) = 92.5$
Rotational Test	$18/20$	$1/20$ (K)	$1/20$ (D)	$0/20$ (F)	$\mathrm{RSI} = 100(0.90 - 0.4(0.05) - 0.6(0.05)) = 85.0$

Table 1: Trial outcomes and performance index calculations for each subsystem (20 trials per test).

Overall, the system consistently achieves guidance, capture, and containment, with remaining limitations primarily related to scaling and durability.

Peg Guidance (GPI = 93.0)	**Chamber Drop (CRI = 92.5)**	**Rotational Test (RSI = 85.0)**
19/20 successes; the single early bypass was due to peg misalignment, which was corrected. No jamming or compression occurred, indicating effective inward channeling.	19/20 successes; one early partial entry occurred before alignment refinement, with full entry achieved thereafter. No bounce-out or missed openings were observed.	18/20 successes; one sticking event (vertical offset) and one base displacement (motor torque) were resolved by adjusting shaft height and securing the base. No incomplete rotations occurred, indicating sufficient torque. Minor instability remains but does not affect operation.

Table 2: Summary of subsystem test performance and key observations.

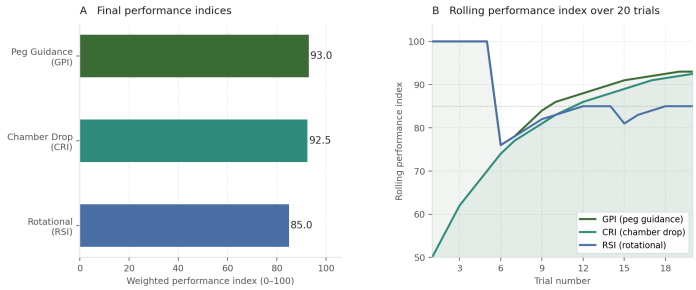

Figure 1: Performance summary across subsystem tests: (A) final indices (all above 80); (B) rolling indices over 20 trials showing convergence of GPI, CRI, and RSI.

Figure 1 summarizes performance progression and final outcomes. Panel A shows all indices exceeding 80, with GPI and CRI highest. Panel B shows GPI and CRI stabilizing quickly above 85, while RSI improves more gradually due to rotational sensitivity. Convergence across all metrics indicates consistent performance rather than isolated favorable trials.

4 Conclusion and Recommendation

The rotating SLF trap demonstrates a practical, non-destructive approach for attracting, guiding, and capturing spotted lanternflies using controlled mechanical motion. The system was evaluated using three success criteria: the Peg Guidance Test (GPI), Chamber Drop Test (CRI), and Rotational System Test (RSI). Across all subsystem tests, the prototype consistently exceeded the minimum performance thresholds, achieving GPI = 93.0, CRI = 92.5, and RSI = 85.0. These results confirm that the system can reliably guide insects toward the center, transfer them into the collection chamber, and operate under continuous rotation with minimal intervention.

Performance improvements observed during testing were primarily driven by small mechanical refinements, including alignment correction and friction reduction. These adjustments significantly reduced bypass, partial entry, and rotational resistance, demonstrating that the design is robust and responsive to incremental optimization. While minor instability remained in the rotational system(due to low infill) it did not prevent successful operation at the prototype scale.

The current prototype shows strong potential but requires scaling and refinement to fully meet our success criteria. To improve the GPI, the system should be built to full scale (1 m height, 1.07 m diameter) with added structural reinforcement, which will reduce deformation and ensure the peg and wall system consistently guide marbles to the center without jamming or escape. Achieving a higher RSI will require a low-speed, high-torque (12V–24V) gear motor and better load distribution, allowing the system to maintain continuous 360° rotation without stalling and ensuring smooth, reliable motion. To enhance the CRI, the last peg on the peg plate should instead be on the reaper wall to allow more room for the SLF to go into the container. Future work should focus on field testing to validate these improvements against the success criteria, ensuring reliability, durability, and efficiency under real-world conditions while refining the design for scalability and cost.

5 Prototyping and Testing Details

The prototype consists of a stationary housing, a peg plate, a rotating reaper wall, a rotating floor, a central opening, and a death chamber located beneath the main trap surface (see Appendix Fig. 2).

To attract spotted lanternflies, tree-of-heaven sap is placed into grooves within the peg plate. As the insects move toward the attractant, the rotating mechanism guides them toward the center of the device, where an opening in the rotating floor allows them to fall into the container below. Once inside the death chamber, the insects are contained until removal. The chamber can be twisted open for easy disposal when full.

The device is assembled in a modular manner. First, the death chamber is secured to the peg plate by rotating it into place. The D-profile shaft is then connected to the motor shaft connector, with the rotating floor attached to the opposite end of the shaft. The shaft assembly is inserted through the motor attachment lid, which is then fastened to the peg plate using bolts. A rectangular housing on top of the lid contains the motor and battery system, completing the assembly.

Appendix A Exploded CAD View

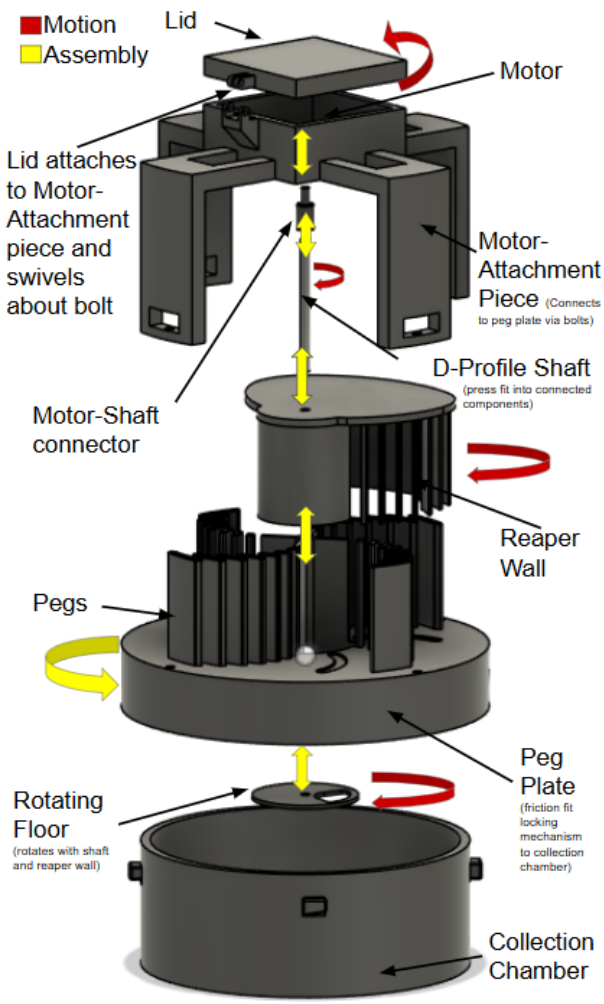

Figure 2: Exploded CAD view showing major components and assembly sequence.

Appendix B Components List and Bill of Materials

Object Description	#	Code / Link	Fabrication Details	Price
D-Profile Rotary Shaft	1	8632T132	Length = 6 in; diameter = 0.25 in; carbon steel; horizontal cut	$7.52
Fielect DC 3V 5RPM Mini Metal Gear Motor GA12-N20	1	Motor link	3V, 5 RPM; length = 27.5 mm / 1.08 in; D-profile shaft	$11.49
Death Chamber	1	N/A	Fly-collecting container; 3D printed by RPL; material: PLA	N/A
Reaper Wall with 6 Pegs and External Wall	1	N/A	3D printed by RPL; material: PLA	N/A
Motor Attachment Lid	1	N/A	3D printed by RPL; material: PLA	N/A
Peg Plate with 28 Pegs	1	N/A	3D printed by RPL; material: PLA	N/A
Motor Shaft Connector	1	N/A	3D printed by RPL; material: PLA	N/A
Rotating Floor	1	N/A	3D printed by RPL; material: PLA	N/A
Lid	1	N/A	3D printed by RPL; material: PLA	N/A
Motor Attachment	1	N/A	3D printed by RPL; material: PLA	N/A
Nuts	1	98676A400	Black-oxide 18-8 stainless steel hex nuts; M6 $\times$ 1 mm	$8.72
Bolts	1	92832A590	M6 zinc-plated steel button head Torx screws	$11.41
POP BABY 50-Piece Multicolor Player Marbles Set	1	Marble link	Diameter: 0.55 in	$4.99
Notches Direct 100,000 100,000 Clear Water Gel Beads for Vases	1	Orbeez link	Diameter: 11–13 mm	$12.99
LAMPVPATH 1.5V AA Battery Holder Case with Wire Leads and ON/Off Switch	1	Battery case link	L $\times$ W $\times$ H: 2.71 in $\times$ 1.34 in $\times$ 0.75 in; wire length: 5.7 in	$4.99

Table 3: Bill of Materials for the prototype.

Appendix C Test Definitions and Performance Metrics

To evaluate prototype performance, three subsystem tests were conducted to assess guidance, capture, and rotational functionality. Each test was evaluated using a weighted performance index that combines outcome frequencies into a single score, with larger penalties assigned to more severe failure modes.

C.1 Test Overview

Test	Part Tested	Metric
Peg Guidance Test	Reaper wall and stationary peg plate	Guidance Performance Index (GPI)
Chamber Drop Test	Center opening and death chamber	Capture Reliability Index (CRI)
Rotational Test with Motor	Full rotating assembly	Rotational Stability Index (RSI)

Table 4: Subsystem tests and corresponding evaluation metrics

C.2 Outcome Legend

The following outcome codes are used in the results table and scoring formulas. Each test uses a distinct set of outcome categories reflecting the specific failure modes relevant to that subsystem.

Test	Code	Outcome	Description
Peg Guidance	S	Success	Marble guided cleanly to center opening
	B	Bypass	Marble exits outer edge without reaching center
	J	Jamming	Marble becomes stuck between pegs mid-path
	C	Compression	Marble is prematurely compressed or crushed by rotating wall
Chamber Drop	S	Success	Marble/Orbeez enters collection chamber fully
	P	Partial entry	Surrogate enters only partway before stopping
	B	Bounce-out	Surrogate enters but rebounds out of chamber
	M	Missed opening	Surrogate misses the chamber opening entirely
Rotational	S	Smooth completion	Full 360° rotation completed without interruption
	K	Sticking	Assembly hesitates or briefly catches during rotation
	D	Base displacement	Motor torque causes the base to shift position
	F	Failed rotation	Assembly cannot complete a full revolution

Table 5: Outcome codes and definitions for all three subsystem tests

C.3 Performance Index Definitions

Peg Guidance Test (GPI). Evaluates the ability of the reaper wall and peg plate to guide surrogate particles toward the central opening.

$$\mathrm{GPI} = 100\,(S_t - 0.4B_t - 0.7J_t - 1.0C_t)$$

Premature compression is assigned the largest penalty as it represents the most severe failure mode.

Chamber Drop Test (CRI). Evaluates alignment between the central opening and the collection chamber based on whether particles successfully enter the chamber.

$$\mathrm{CRI} = 100\,(S_c - 0.5P_c - 0.7B_c - 1.0M_c)$$

Missed openings receive the largest penalty as they indicate significant misalignment.

Rotational Test with Motor (RSI). Evaluates the ability of the rotating assembly to complete a full revolution under motor power.

$$\mathrm{RSI} = 100\,(S_r - 0.4K_r - 0.6D_r - 1.0F_r)$$

Incomplete rotation is penalized most as it indicates insufficient motor torque.